

TANGIER IBN BATTUTA, Morocco

WMO: 601010

Lat: 35.727N

Lon: 5.917W

Elev: 19

StdP: 101.10

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.2	5.8	0.5	3.9	8.9	2.1	4.4	9.1	14.0	14.4	12.8	14.5	2.0	90	0.496

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.0	33.2	21.3	31.9	21.2	30.2	20.9	23.0	29.9	22.5	28.9	22.0	27.9	8.4	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	15.7	25.8	20.2	14.9	25.3	19.9	14.7	25.1	68.2	30.2	66.1	28.8	64.0	27.9	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.9	12.4	11.1	1.6	37.4	1.9	2.2	0.3	39.0	-0.8	40.3	-1.9	41.5	-3.3	43.1		
			0.9	24.0	1.9	0.7	-0.5	24.5	-1.6	24.9	-2.6	25.3	-4.0	25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.3	12.6	12.9	14.7	16.2	18.9	22.2	24.0	24.8	22.9	20.6	16.2	13.9
	DBStd	4.96	2.23	2.36	2.29	2.28	2.69	2.77	2.60	2.67	2.50	2.50	2.77	2.29
	HDD10.0	11	4	4	1	0	0	0	0	0	0	0	1	2
	HDD18.3	763	178	151	114	71	25	1	0	0	0	8	75	139
	CDD10.0	3059	84	87	146	187	274	366	433	459	387	328	186	122
	CDD18.3	768	0	0	1	8	41	117	175	201	137	77	11	0
	CDH23.3	5296	0	1	11	32	241	809	1407	1693	804	283	15	0
	CDH26.7	1662	0	0	2	2	48	259	502	605	203	40	1	0
Wind	WSAvg	5.0	4.5	5.0	5.4	5.3	5.1	5.2	4.8	4.9	5.0	4.8	4.7	4.7
Precipitation	PrecAvg	669	91	69	84	49	35	7	1	2	27	83	110	110
	PrecMax	1574	473	279	232	149	115	67	10	13	63	218	298	471
	PrecMin	288	0	0	18	0	2	0	0	0	0	2	8	0
	PrecStd	249	92	61	47	33	30	14	3	3	17	57	81	107
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.9	21.1	24.5	26.1	30.1	34.2	35.2	36.0	32.6	29.2	25.1	21.2
		MCWB	15.0	15.4	16.7	17.1	18.4	19.8	21.2	21.4	20.9	19.4	17.6	15.1
	2%	DB	18.4	19.2	21.9	24.1	27.9	31.7	33.1	33.2	30.7	27.8	23.1	19.8
		MCWB	14.4	14.8	15.6	16.4	18.0	19.6	21.3	22.0	21.3	19.6	17.3	15.2
	5%	DB	17.7	18.1	20.2	22.3	26.0	29.7	31.3	31.9	29.1	26.1	21.9	18.8
		MCWB	14.2	14.3	14.9	16.0	17.6	19.6	21.2	21.9	20.8	19.2	17.1	15.0
	10%	DB	16.8	17.1	19.0	21.0	24.1	27.8	29.8	30.2	27.2	24.9	20.5	17.8
		MCWB	13.7	13.9	14.6	15.8	17.4	19.2	21.0	21.7	20.4	19.1	16.5	14.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.6	16.7	18.0	18.7	20.5	22.1	23.6	24.0	22.7	21.6	19.8	17.7
		MCDB	18.2	19.0	22.7	23.7	26.6	29.7	31.5	31.6	29.4	25.9	22.0	19.1
	2%	WB	15.5	15.7	16.6	17.7	19.3	21.2	22.7	23.2	22.2	20.7	18.5	16.6
		MCDB	17.5	18.1	19.9	22.1	24.9	27.9	29.8	30.1	28.1	25.1	21.6	18.3
	5%	WB	14.7	15.0	15.8	17.0	18.6	20.7	22.2	22.7	21.5	20.1	17.6	15.7
		MCDB	16.8	17.5	18.9	20.9	23.9	26.8	28.7	29.1	26.9	24.4	20.8	17.8
	10%	WB	14.0	14.2	15.2	16.2	17.9	20.1	21.5	22.2	21.0	19.5	16.8	14.9
		MCDB	16.3	16.7	18.3	19.9	22.9	25.6	27.5	28.1	25.8	23.6	19.9	17.3
Mean Daily Temperature Range	5% DB	MDBR	8.0	8.1	8.3	8.4	9.1	9.1	9.3	9.0	8.2	7.9	7.5	7.5
		MCDBR	8.7	9.0	10.5	10.7	11.7	11.9	12.4	11.4	9.8	9.3	8.4	8.2
		MCWBR	5.9	6.1	6.3	6.1	5.6	4.7	4.8	4.3	4.0	4.2	4.8	5.3
	5% WB	MCDBR	7.6	8.1	9.3	9.6	10.6	10.3	10.8	10.0	8.6	8.2	7.3	6.9
		MCWBR	5.8	6.0	6.3	6.2	5.8	4.8	5.0	4.5	4.1	4.5	4.8	5.3
Clear-Sky Solar Irradiance	taub	0.337	0.362	0.406	0.402	0.418	0.441	0.448	0.458	0.430	0.398	0.364	0.343	
	taud	2.506	2.429	2.285	2.327	2.309	2.253	2.222	2.202	2.314	2.404	2.473	2.504	
	Ebn at Noon	860	875	860	883	869	845	836	821	827	823	820	828	
	Edh at Noon	83	100	125	126	130	137	141	141	120	101	85	79	
All-Sky Solar Radiation	RadAvg	2.67	3.50	4.61	5.79	6.77	7.60	7.68	6.79	5.37	3.92	2.85	2.38	
	RadStd	0.25	0.40	0.45	0.33	0.44	0.31	0.16	0.22	0.24	0.34	0.23	0.22	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only		N/A	N/A	N/A	N/A	N/A	-0.36	N/A	N/A	N/A	+109
Regional (8 neighbors)		+0.19	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+57

Nomenclature: See separate page